SOFTWARE ENGINEERING LAB-01 (UE24CS341A)

TEAM NO – 001

PROBLEM STATEMENT – 001
Smart Lab Equipment Slot Reservation Portal

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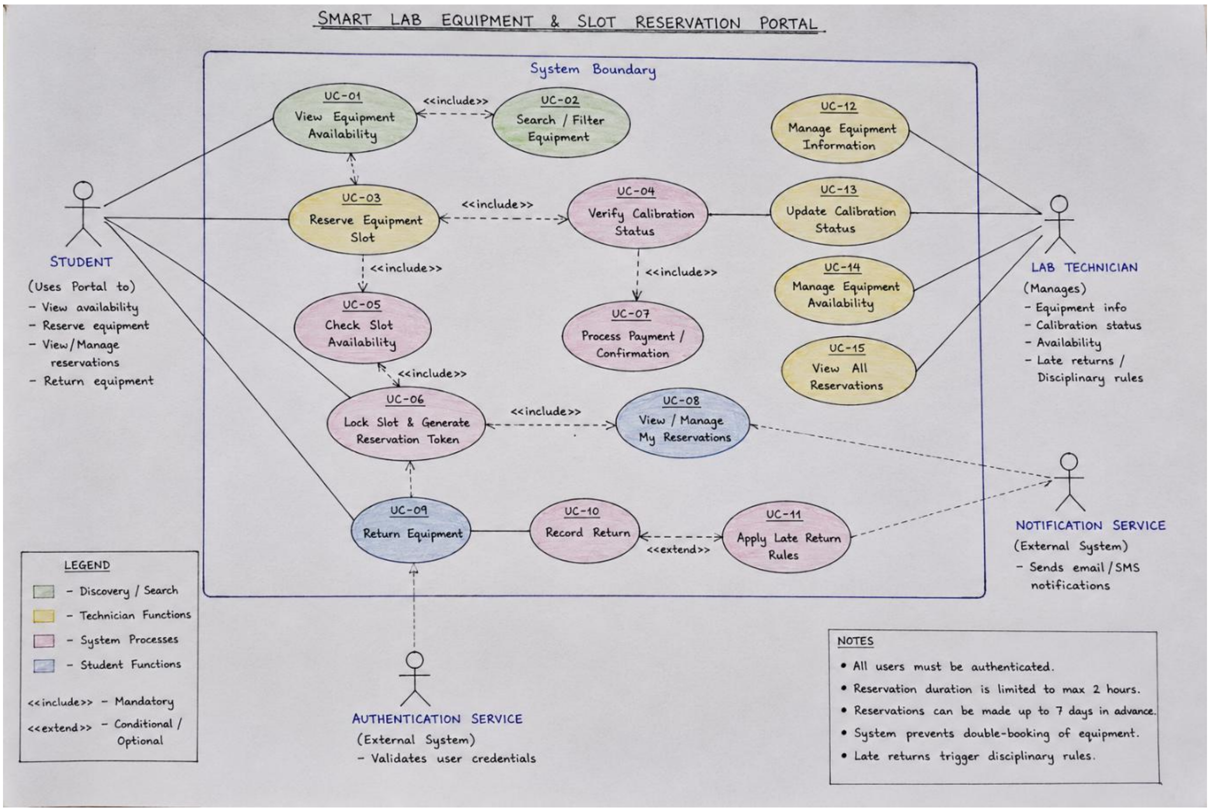
1. Requirements Table

Five Functional Requirements (FR-001–FR-005, with FR-001 as given) and two Nonfunctional Requirements (NFR-001–NFR-002, with NFR-001 as given).

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow authenticated students to view real-time availability and reserve a maximum 2-hour slot for any calibrated equipment up to 7 days in advance.	High	Pass: The slot is successfully locked and a reservation token is generated. Fail: Double-booking is allowed or the reservation exceeds the 2-hour limit.	Core functionality for reserving high-value lab equipment while preventing scheduling conflicts.
FR-002	Functional	The system shall verify the calibration status of selected equipment before confirming a reservation.	High	Pass: A reservation is confirmed only when the selected equipment has valid calibration status. Fail: The system confirms a reservation for equipment whose calibration is invalid or expired.	Ensures students reserve equipment that is currently calibrated and suitable for use.
FR-003	Functional	The system shall prevent two students from reserving the same equipment for overlapping time slots.	High	Pass: When two requests overlap, only one reservation is confirmed and the other request is rejected or redirected to an available slot. Fail: Both overlapping reservations are confirmed.	Prevents double-booking of high-value laboratory equipment.
FR-004	Functional	The system shall allow students to return reserved equipment and record whether the equipment was returned after the reserved slot.	Medium	Pass: A return is recorded with its return time and a late return is identified when applicable. Fail: The system cannot record the return or fails to identify a late return.	Supports equipment tracking and enforcement of late-return rules.
FR-005	Functional	The system shall allow lab technicians to update equipment availability and calibration status.	High	Pass: An authorized lab technician can update an equipment status and the updated status is reflected in the portal. Fail: An unauthorized user can make the update or the updated status is not reflected.	Keeps equipment availability and calibration information accurate for reservations.
NFR-001	Performance & Security	The system shall ensure that concurrent slot reservation requests are processed within 200 ms with strict database-level lock isolation to prevent race conditions.	High	Pass: Benchmarking under simulated peak load confirms the target latency of 200 ms or less and no conflicting reservations are created. Fail: Requests exceed the target latency or race conditions cause conflicting reservations.	Ensures fast, secure, and consistent reservation processing when multiple students attempt to book simultaneously.
NFR-002	Reliability	The system shall preserve confirmed reservation records and reservation tokens without data loss when a reservation transaction is successfully completed.	High	Pass: After a successful reservation, the reservation record and token remain retrievable after logout, refresh, or system restart. Fail: A successfully confirmed reservation or its token is lost.	Ensures confirmed bookings remain reliable and traceable for students and lab technicians.

2. UML Use-Case Diagram



Include relationships:

"View Equipment Availability" includes "Search / Filter Equipment"; "Reserve Equipment Slot" includes "Verify Calibration Status" and "Check Slot Availability"; "Reserve Equipment Slot" includes "Lock Slot & Generate Reservation Token"; "Verify Calibration Status" includes "Process Payment / Confirmation"; "Lock Slot & Generate Reservation Token" includes "View / Manage My Reservations"; "Return Equipment" includes "Record Return".

Extend relationships:

"Manage Return & Late-Return Rules" extends "Record Return" (conditional, only when the equipment is returned late).

3. Use-Case Flow Specification

Use Case: Check Slot Availability

Primary Actor: Student
Secondary Actor: Lab Technician

Precondition: Student is authenticated and has access to the Smart Lab Equipment & Slot Reservation Portal. The selected equipment exists in the system and its reservation information is available.

Postcondition (Success): The system displays whether the selected equipment and requested time slot are available for reservation.

Postcondition (Failure): The system informs the student that the selected slot is unavailable and displays alternative available slots, if any.

Main Success Scenario

1. Student selects **“Check Slot Availability”** from the equipment reservation interface.
2. System displays the selected equipment and the available date and time slots.
3. Student selects the required date and time slot.
4. System retrieves the existing reservations for the selected equipment.
5. System checks whether another reservation overlaps with the requested time slot.
6. System confirms that the requested slot is available.
7. System marks the selected slot as **Available**.
8. System displays the availability status to the student.
9. Student can proceed with the **“Reserve Equipment Slot”** use case.
10. Use case ends successfully.

Alternate Flow

5a. Selected Slot Is Already Reserved

- **5a1.** System detects an existing reservation that overlaps with the selected time slot.
- **5a2.** System marks the selected slot as **Unavailable**.
- **5a3.** System informs the student that the selected equipment is already reserved for the requested time.
- **5a4.** System displays other available time slots for the selected equipment.
- **5a5.** Student selects an alternative available slot.
- **5a6.** System checks the availability of the newly selected slot.
- **5a7.** If the new slot is available, the system displays it as **Available** and the student can proceed with the reservation.

- **5a8.** If the student does not select another slot, the use case ends without creating a reservation.

This one is **very safe for your assignment** because the central behaviour is simply checking whether a slot is already reserved, which directly supports the problem statement's requirement to prevent double-booking.